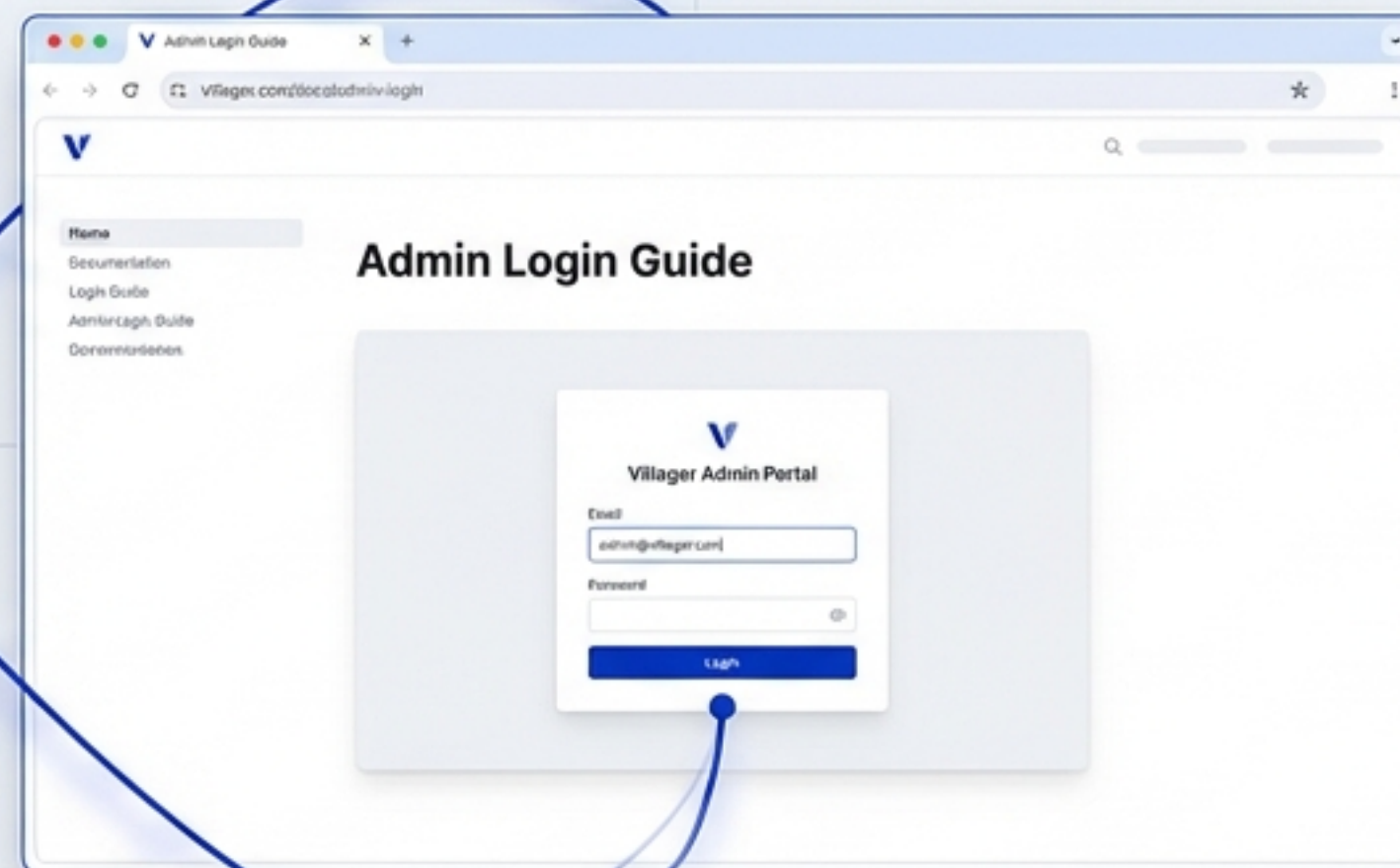


Project Villager

Browser Automation & Living Documentation

A dual-purpose infrastructure for UI validation and automated user guides.

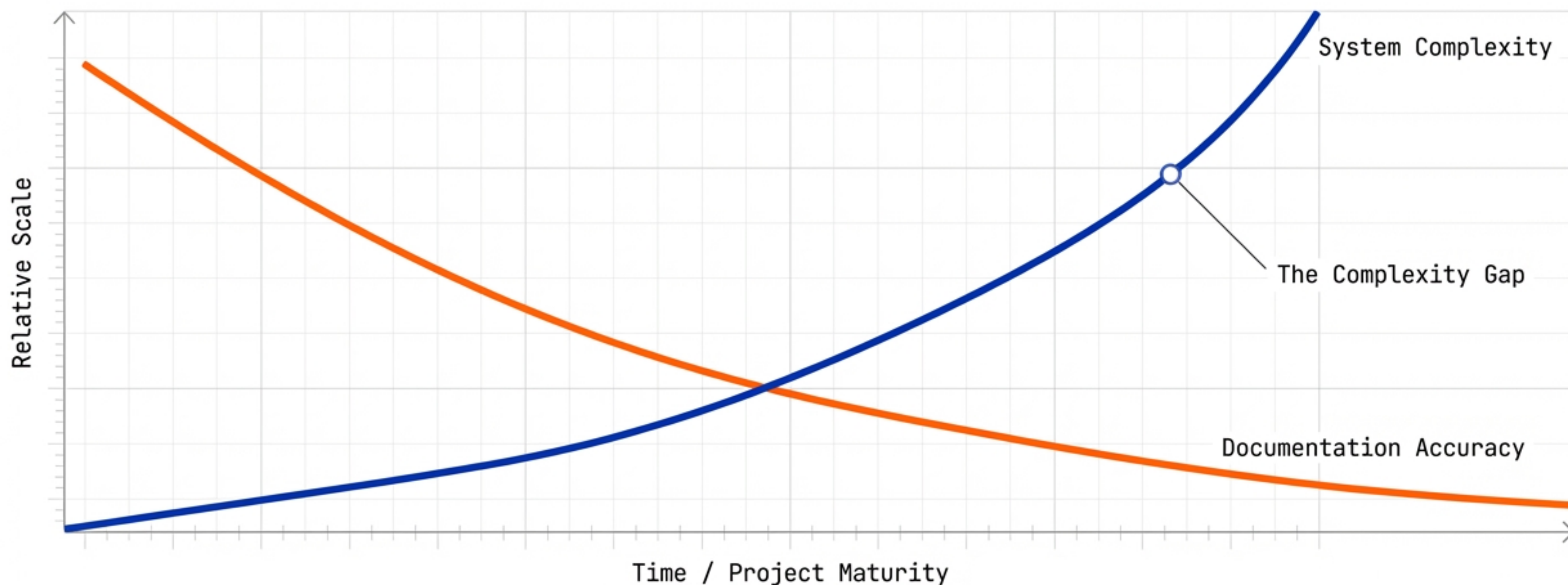
```
def test_admin_login(browser):  
    browser.goto('/admin')  
    browser.fill('#email', 'admin@villager.com')  
    browser.screenshot(path='docs/assets/login.png')
```



The Efficiency Paradox

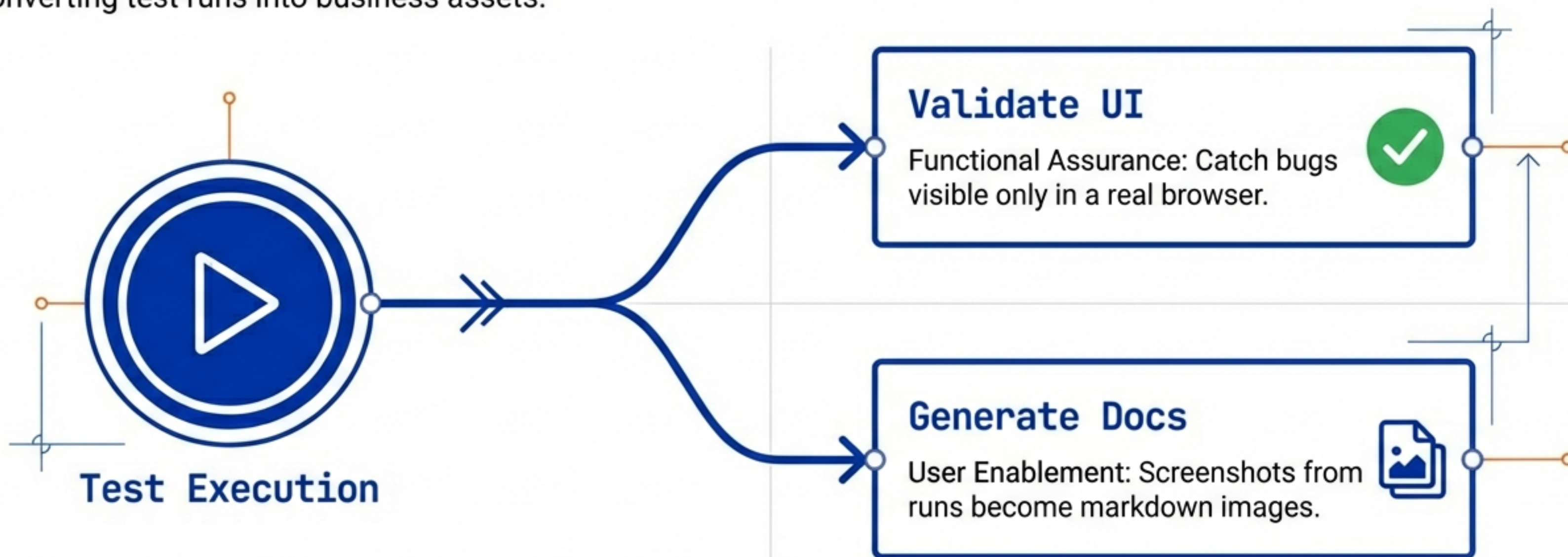
As workflows like PKI key management and fleet management grow complex, manual testing fails to scale. Simultaneously, static documentation drifts from reality.

Automated tests usually validate code without producing anything humans can read. **We need a solution where the test IS the documentation.**



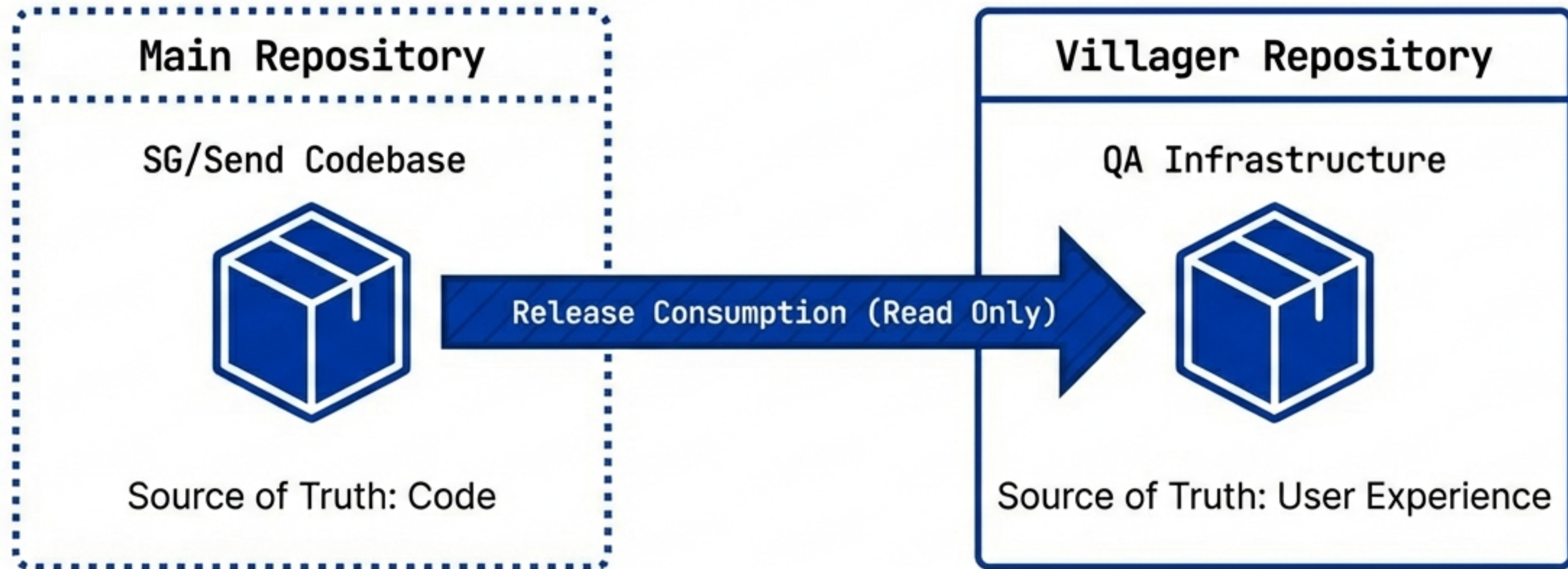
Two Birds, One Stone

Converting test runs into business assets.



**Every test run proves the feature works
AND explains how it works.**

Architectural Separation: The Villager Model



Formal QA

Consumes releases,
not dev branches.

Clean Separation

Keeps binary data
(screenshots) out of source.

Independence

Decoupled release cadence.

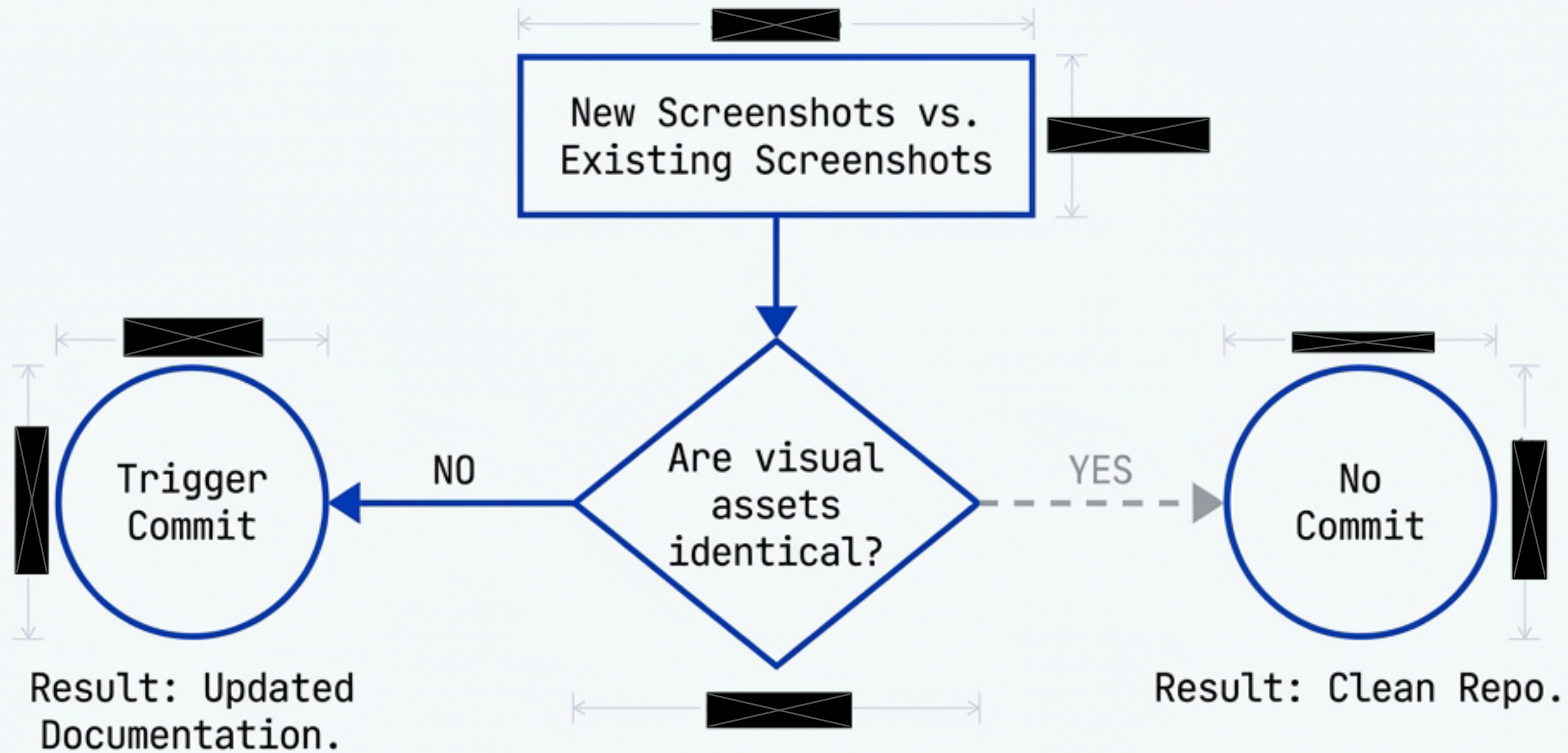
The Execution Flow: Capture

Phase 1: Execution Flow



NOTE: The suite starts by downloading the release and running it locally.

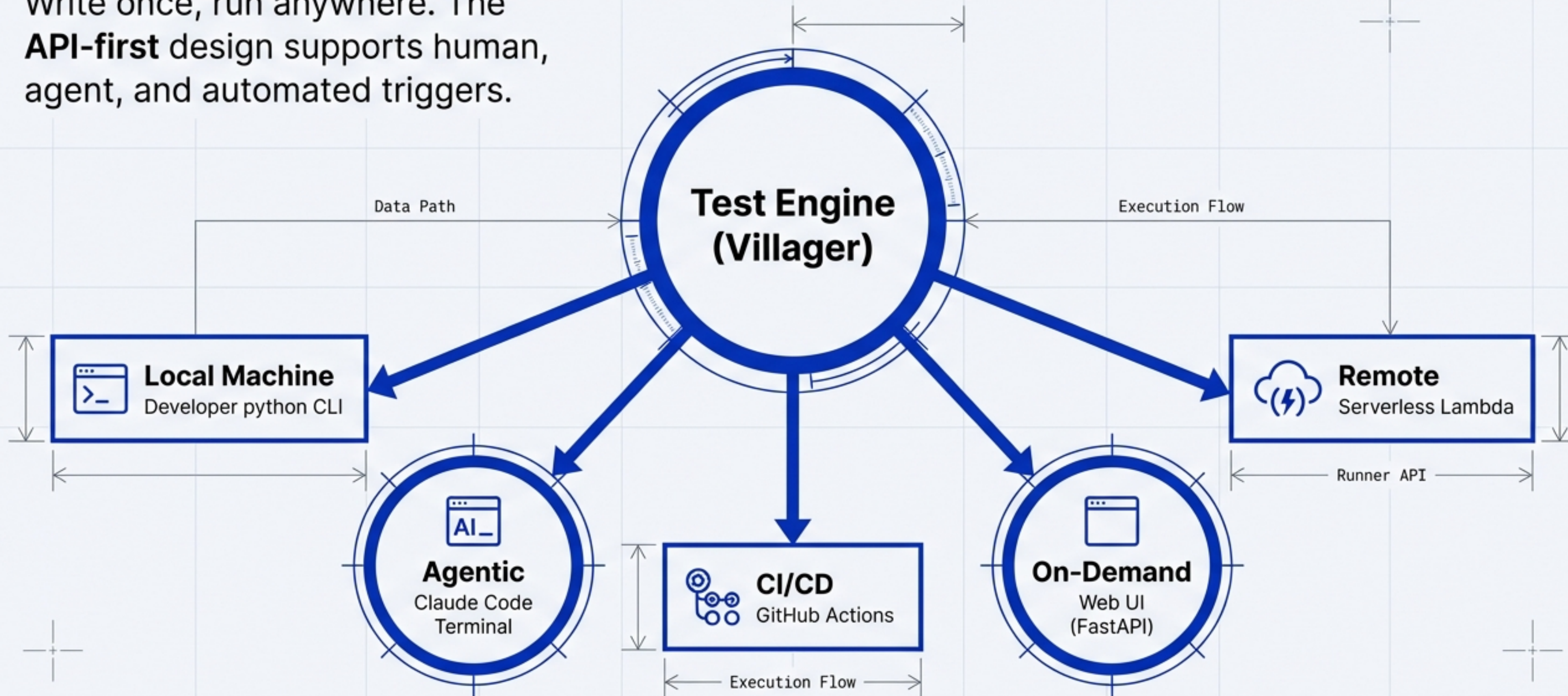
The Logic Core: Visual Diffing



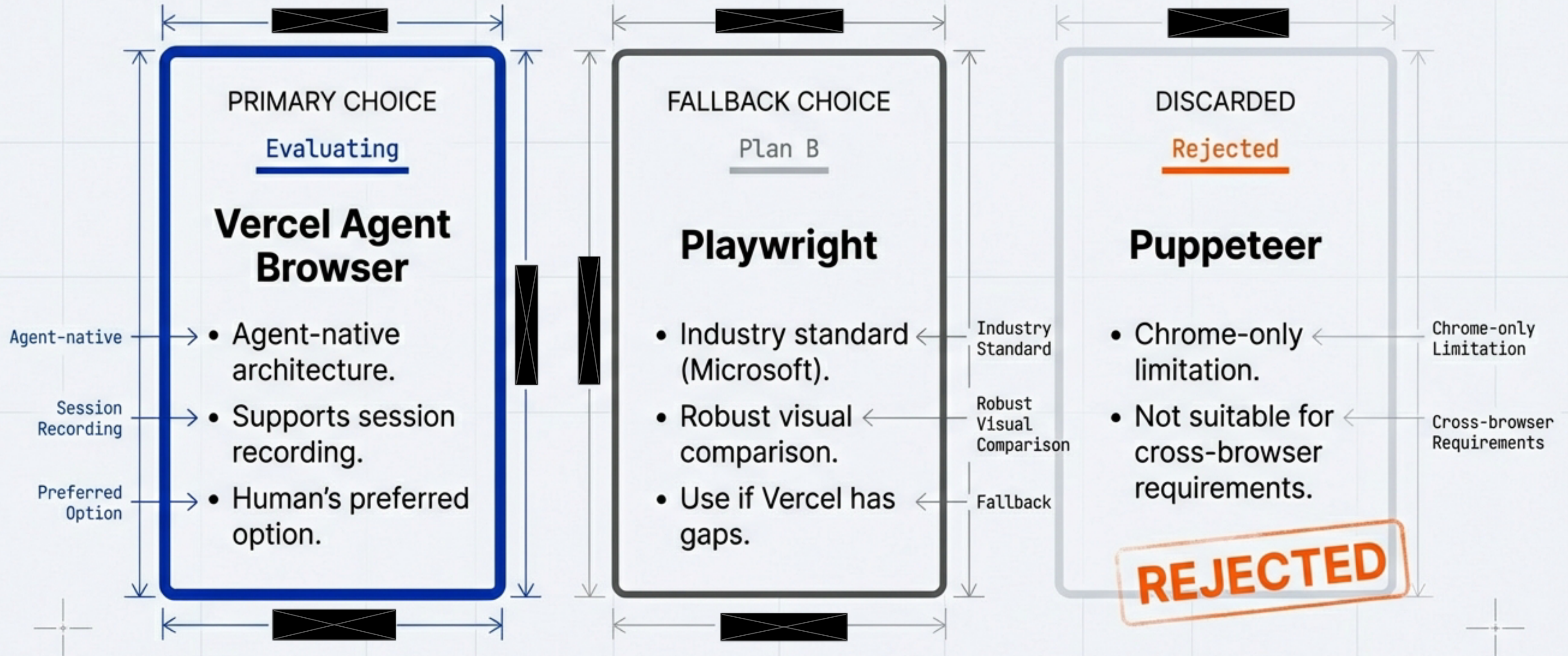
The git diff history becomes a strictly visual history of UI evolution.

One Suite, Many Runners

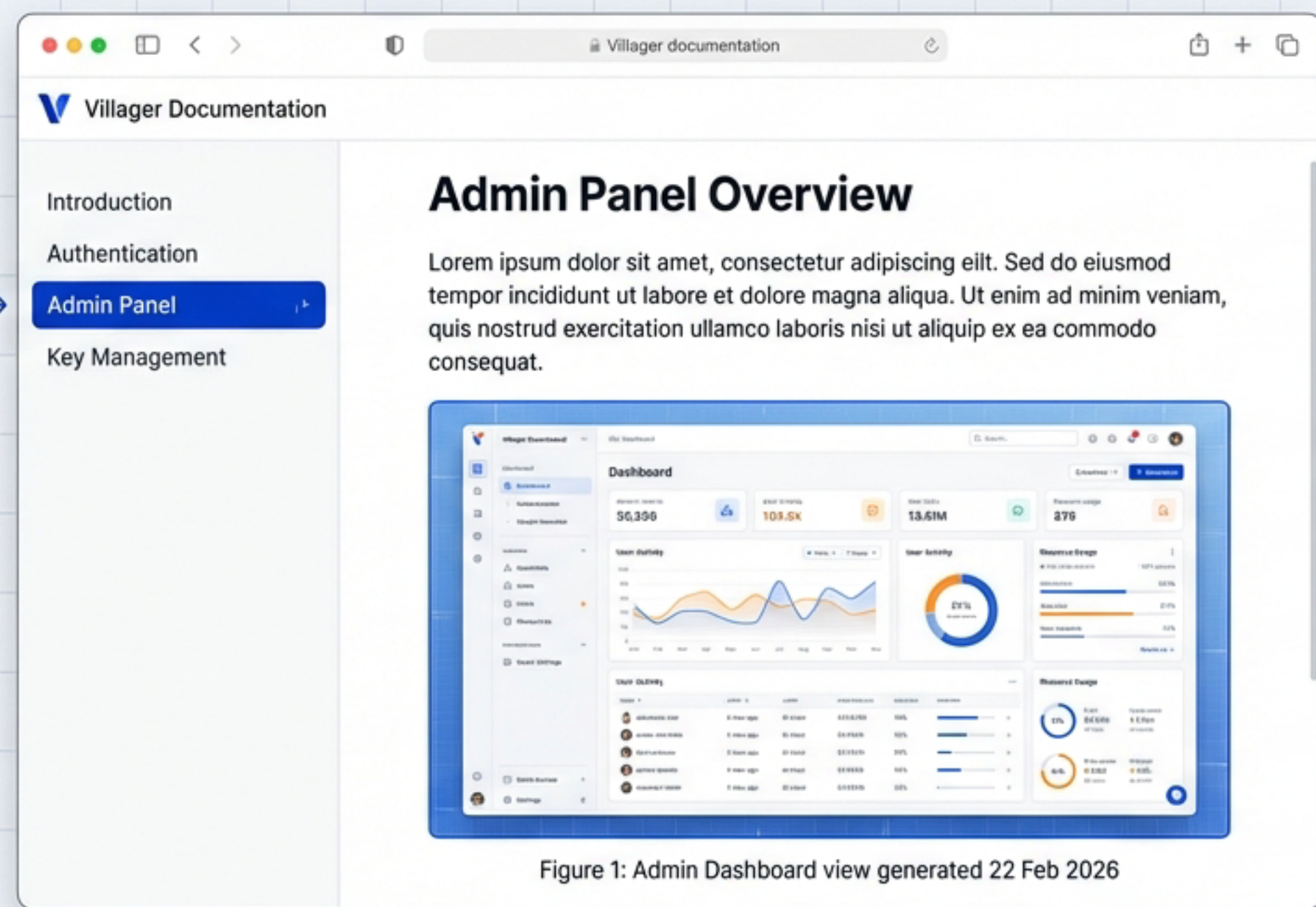
Write once, run anywhere. The **API-first** design supports human, agent, and automated triggers.



Tooling Strategy: Agent-First Automation



The Output: Automated GitHub Pages



- ✓ Zero manual documentation effort
- ✓ No stale screenshots
- ✓ No documentation drift
- ✓ Automated GitHub Pages deployment

Role Allocation



QA (Lead)

Test design & Quality gates.



Developer

Implementation & Runner



Architect

Strategy & Tooling.



Sherpa

User-facing text translation.



Librarian

Cataloguing & Context.



DevOps

CI & Deployments.

Acceptance Criteria

The Definition of Done

Infrastructure

- ☐ Separate 'Villager' repo created
- ☐ CI/CD Pipeline (GitHub Actions) active
- ☐ Published to GitHub Pages

Functionality

- ☐ Navigate Admin Home & enter Access Token
- ☐ Smart Visual Diff (No change = No commit)
- ☐ Execution via CLI, API, and Web UI

CLASSIFICATION: D1

Infrastructure / Enabler

Mobilisation

**Architect &
Developer**

Begin evaluation
of Vercel Agent
Browser
immediately. 

DevOps

Initialize the
separate Villager
repository.  

0%

Constructing the Living
Documentation System. 

